SUMS OF POWERS OF INTEGERS: A COMPLETE FRAMEWORK FOR CLOSED FORMULAS

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Abstract. This manuscript originates from the historical development of the sums of powers problem, briefly tracing its evolution from Archimedes (287-212 BC) to Donald Knuth's work *Johann Faulhaber and sums of powers (1993)*, highlighting the transition from purely arithmetic methods to modern enumerative and algebraic-combinatorial approaches. The main results establish an algorithm for deriving closed formulas for multifold sums of powers of integers by combining variations of Newton's interpolation formula with hockey-stick family identities for binomial coefficients.

Contents

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1. Historical context

The story begins in ancient Greek times, with discrete approximations to obtain continuous areas by the method of exhaustion. In particular, Archimedes (287-212 BC) evaluated sums of squares while finding the spiral's area [1], analogous to what we have nowadays

$$1^2 + 2^2 + 3^2 + \cdots n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Moving to early modern Europe (1636), Pierre de Fermat (1601-1665) referred to the problem of finding formulas for sums of powers ... *perhaps the most beautiful problem of all arithmetics*, in his correspondence with Blaise Pascal (1623-1662). Fermat's contribution to the topic was quite remarkable indeed, as he revived the study of figurate numbers and investigated their arithmetic properties, including what is now known as Fermat's polygonal number theorem. Application of polygonal numbers in sums of powers is extensively discussed in [2, (2020)].

Meanwhile, Johann Faulhaber (1580-1635) worked on the problem of sums of powers, writing his famous book *Academia Algebrae* [3, (1631)]. Faulhaber's contribution was great indeed, because Johann managed to compute formulas up to $\Sigma^1 n^{25} = 1^{25} + 2^{25} + \cdots + n^{25}$, which is quite impressive. Imagine to derive the formula back in 17-th century, without computers. Faulhaber explicitly tabulated formulas through the seventeenth power and provided sufficient information for determining higher cases, including the twenty-fifth power. Knuth later reconstructed the latter computation. Faulhaber actually discovered deep structural patterns, particularly for odd powers expressed as polynomials in triangular numbers. What he lacked was a completely general symbolic formula.

The major breakthrough was achieved by Jacob Bernoulli (1665-1705) in his work entitled *Ars coniectandi* [4, (1713)], published postmortem. In the work, Jacob introduced Bernoulli numbers (named after him later), thus allowing for arbitrary powers to be summed up

$$\sum_{k=1}^{n} k^p = \frac{1}{p+1} \sum_{r=0}^{p} \binom{p+1}{r} B_r n^{p+1-r}$$

It assumes that $B_1 = +\frac{1}{2}$. The choice of convention for B_1 has a long historical background; see [5, 6] for a detailed discussion. Therefore, Bernoulli managed to contribute the missing capstone to Faulhaber's work, making it symbolically complete. In fact, the formula above is commonly referred to as Faulhaber's formula, although the modern Bernoulli-number formulation originates with Bernoulli's great work.

During the eighteenth and nineteenth centuries, the theory was further developed by Euler, Stirling, Jacobi, and others through connections with finite differences, interpolation, and special number sequences.

In 1993, Donald Knuth revisited the topic of sums of powers, giving it a new direction of development. Based on works of Carl Gustav Jacobi (1804-1851), and Gessel-Viennot [7, (1989)], Knuth reviewed Faulhaber's work from a completely different angle, by using modern enumerative combinatorics techniques. Remarkably, Knuth's contribution does not necessarily involve Bernoulli numbers, instead he derived formulas for power sums by utilizing special numbers (e.g Stirling, central factorial), and Binomial coefficients. In addition, Knuth introduced a remarkably convenient notation for r-fold sums of powers $\Sigma^r n^m$, which we use in this work.

In this manuscript, we reveal the role of Newton's interpolation formula in the process of computing formulas for sums of powers. As the main result, this paper proposes an algorithm to derive closed formulas for sums of powers, by utilizing variations of Newton's interpolation formula, combined with hockey-stick family binomial identities. As results show, there are infinitely many variations of closed formulas for sums of powers.

2. Definitions and notations

Allow us to define the mathematical objects we utilize throughout the manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [8], because it is simple and beautiful.

Definition 2.1 (Multifold sums of powers)**.** *For non-negative integers r, n, m*

$$\begin{aligned}\Sigma^0 n^m &= n^m,\\ \Sigma^1 n^m &= \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,\\ \Sigma^{r+1} n^m &= \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.\end{aligned}$$

We utilize the following notation for falling factorials

Definition 2.2 (Falling factorials)**.** *For integers x, n*

$$(x)_n = \begin{cases} 1, & \text{if } k = 0,\\ x(x-1)(x-2)\cdots(x-n+1) = \prod_{k=0}^{n-1}(x-k), & \text{if } k > 0,\\ 0, & \text{else.}\end{cases}$$

We utilize the following notation for central factorials

Definition 2.3 (Central factorials)**.** *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0,\\ 1, & \text{if } k = 0,\\ n\left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = n\left(n+\frac{k}{2}-1\right)_{k-1}, & \text{if } k > 0,\end{cases}$$

where $\left(n+\frac{k}{2}-1\right)_{k-1} = \prod_{j=0}^{k-2}\left(n+\frac{k}{2}-1-j\right)$ are falling factorials.

We encourage the audience to read J. F. Steffensen's work on the definition of generalized factorials [9].

3. Auxiliary lemmas

Lemma 3.1 (Central Newton's formula). *For a function f defined on integers and arbitrary integers x, t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where $\delta^k f(t)$ denotes the k-th central finite difference

$$\delta^k f(t) = \sum_{j=0}^{k} (-1)^j \binom{k}{j} f\left(t + \tfrac{k}{2} - j\right),$$

and $(x-t)^{[k]}$ are central factorials defined by (2.3).

Proof. See [10, p. 462], and [11, section 2, eq. (19)], and [12, section 6, eq. (21)]. □

Lemma 3.2 (Newton's formula for powers). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$n^m = \sum_{k=0}^{m} \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

Lemma 3.3 (Central factorials binomial form). *For integers n and $k \geq 1$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k(k-1)!} \left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$. □

Lemma 3.4 (Central factorials binomial form decomposition). *For integers $n, k \geq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \left[\binom{n + \frac{k}{2}}{k} + \binom{n + \frac{k}{2} - 1}{k} \right].$$

Proof. We have $\binom{a}{k} = \frac{a}{k} \binom{a-1}{k-1}$. Let $a = n + \frac{k}{2}$, then

$$\tbinom{n+\frac{k}{2}}{k} = \tfrac{n+\frac{k}{2}}{k} \tbinom{n+\frac{k}{2}-1}{k-1} = \tfrac{n}{k} \tbinom{n+\frac{k}{2}-1}{k-1} + \tfrac{1}{2} \tbinom{n+\frac{k}{2}-1}{k-1}.$$

Thus,

$$\tfrac{n}{k}\tbinom{n+\frac{k}{2}-1}{k-1} = \tbinom{n+\frac{k}{2}}{k} - \tfrac{1}{2}\tbinom{n+\frac{k}{2}-1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\tfrac{n}{k}\tbinom{n+\frac{k}{2}-1}{k-1} = \tfrac{1}{2}\left[2\tbinom{n+\frac{k}{2}}{k} - \tbinom{n+\frac{k}{2}-1}{k-1}\right] = \tfrac{1}{2}\left[\tbinom{n+\frac{k}{2}}{k} + \tbinom{n+\frac{k}{2}-1}{k} + \tbinom{n+\frac{k}{2}-1}{k-1} - \tbinom{n+\frac{k}{2}-1}{k-1}\right].$$

Thus, the statement follows. □

Lemma 3.5 (Newton's formula for powers decomposition)**.** *For integers* $n, m \geq 0$*, and an arbitrary integer* t*,*

$$n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}}{k} + \binom{n-t+\frac{k}{2}-1}{k}\right]\delta^k t^m.$$

Lemma 3.6 (Generalized hockey-stick identity)**.** *For integers* a, b *and* j*,*

$$\sum_{k=a}^{b}\binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^{b}\binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^{b}\binom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\binom{k}{j}\right)$. By hockey-stick identity $\sum_{k=0}^{n}\binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^{b}\tbinom{k}{j} = \left(\sum_{k=0}^{b}\tbinom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\tbinom{k}{j}\right) = \tbinom{b+1}{j+1} - \tbinom{a}{j+1}.$$

This completes the proof. □

Lemma 3.7 (Centered hockey-stick identity)**.** *For integers* $n \geq 1$*, and arbitrary integers* t, k, r

$$\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Proof. By setting $a = 1-t+\frac{k}{2}+r$, and $b = n-t+\frac{k}{2}+r$ into Lemma (3.6) yields

$$\sum_{j=1}^{n}\tbinom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r}\tbinom{j}{k} = \tbinom{n-t+\frac{k}{2}+r+1}{k+1} - \tbinom{1-t+\frac{k}{2}+r}{k+1}.$$

This completes the proof. □

Convention 3.8. *For all x*

$$x^0 = 1.$$

Donald Knuth give extensive discussion on the convention $x^0 = 1$ for all x in Concrete Mathematics [13, p. 162], and in [14].

4. Ordinary sums of powers

By decomposition of Newton's formula for powers (3.5), for integers $n, m \geq 0$, and an arbitrary integer t, we get

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^{m} \left[\sum_{j=1}^{n} \binom{j - t + \frac{k}{2}}{k} + \sum_{j=1}^{n} \binom{j - t + \frac{k}{2} - 1}{k} \right] \delta^k t^m.$$

Now we reach the transition to closed form of ordinary sums of powers. By setting $r = 0$, and $r = -1$ into Lemma (3.7), we get closed forms of the sums $\sum_{j=1}^{n} \binom{j-t+\frac{k}{2}}{k}$, and $\sum_{j=1}^{n} \binom{j-t+\frac{k}{2}-1}{k}$, respectively. Therefore,

Proposition 4.1 (Ordinary sums of powers decomposition)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^{m} \left[\binom{n - t + \frac{k}{2} + 1}{k + 1} - \binom{1 - t + \frac{k}{2}}{k + 1} + \binom{n - t + \frac{k}{2}}{k + 1} - \binom{-t + \frac{k}{2}}{k + 1} \right] \delta^k t^m.$$

Note that we should note worry about fractions in upper index of binomial coefficients $\binom{n-t+\frac{k}{2}+1}{k+1}$, because binomial coefficient $\binom{n}{k}$ is a polynomial in n. The well-known identity in unsigned Stirling numbers of the first kind $\left[{n \atop k}\right]$ shows it clearly

$$\binom{n}{k} = \sum_{j=0}^{k} \frac{(-1)^{k-j}}{k!} \left[{k \atop j}\right] n^j.$$

For example, the sums of squares are

Example 4.2 (Sums of squares in central differences).

$$
\begin{aligned}
t=0: &\quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(1+n) + \frac{\delta^2 t^2}{2}\, \frac{n(1+3n+2n^2)}{6},\\
t=1: &\quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(n-1) + \frac{\delta^2 t^2}{2}\, \frac{n(1-3n+2n^2)}{6},\\
t=2: &\quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(n-3) + \frac{\delta^2 t^2}{2}\, \frac{n(13-9n+2n^2)}{6},\\
t=3: &\quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(n-5) + \frac{\delta^2 t^2}{2}\, \frac{n(37-15n+2n^2)}{6},\\
t=4: &\quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(n-7) + \frac{\delta^2 t^2}{2}\, \frac{n(73-21n+2n^2)}{6}.
\end{aligned}
$$

While the sums of cubes

Example 4.3 (Sums of cubes in central differences).

$$
\begin{aligned}
t=0: &\quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(1+n) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(1+3n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-1+n+4n^2+2n^3)}{24},\\
t=1: &\quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-1) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(1-3n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(1+n-4n^2+2n^3)}{24},\\
t=2: &\quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-3) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(13-9n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-21+25n-12n^2+2n^3)}{24},\\
t=3: &\quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-5) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(37-15n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-115+73n-20n^2+2n^3)}{24},\\
t=4: &\quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-7) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(73-21n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-329+145n-28n^2+2n^3)}{24}.
\end{aligned}
$$

It is indeed interesting to notice the connection between the decomposition of ordinary sums of powers (4.1) and Riordan's central factorial numbers of the second kind, that is

Proposition 4.4 (Central factorial numbers sums). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$
\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n+\frac{k}{2}+1}{k+1} - \binom{1+\frac{k}{2}}{k+1} + \binom{n+\frac{k}{2}}{k+1} - \binom{\frac{k}{2}}{k+1}\right] k!T(m,k),
$$

where $T(m,k)$ denote the central factorial numbers of the second kind (in Riordan's notation). See [12, p. 213], *and* [15].

These central factorial numbers of the second kind $T(n,k)$ are defined by central Newton's formula for powers (3.2) evaluated in zero, such that

$$n^m = \sum_{k=0}^{\infty} T(m,k) n^{[k]}.$$

Which implies

$$T(n,k) = \frac{\delta^k 0^m}{k!},$$

by Newton's formula (3.2).

5. Remark on Knuth's formula for sums of odd powers

In [8] Donald Knuth gives a remarkable formula for sums of odd powers

Proposition 5.1 (Knuth's odd powers sum)**.** *For non-negative integers n, m*

$$\Sigma^1 n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k}{2k} T(2m, 2k),$$

where $T(2m,2k)$ are central factorial numbers of the second kind.

It is quite interesting to notice that Proposition (5.1) originates from centered Newton's formula for powers (3.2) evaluated at zero. By Newton's formula, we have

$$n^{2m} = \sum_{k=0}^{2m} \frac{n^{[k]}}{k!} \cdot \delta^k 0^{2m}$$

The iteration for $k=0$ can be omitted because $\delta^0 0^{2m} = 0$, for integer $m \geq 1$. Next, we observe that $\delta^k 0^{2m} = 0$ for every odd k, thus we can omit odd iterations of k

$$n^{2m} = \sum_{k=1}^{m} \frac{n^{[2k]}}{(2k)!} \cdot \delta^{2k} 0^{2m}.$$

By definition of central factorial numbers of the second kind, yields

$$n^{2m} = \sum_{k=1}^{m} n^{[2k]} T(2m, 2k).$$

Thus,

$$n^{2m} = \sum_{k=1}^{m} (2k)! \frac{n^{[2k]}}{(2k)!} T(2m, 2k)$$

By simplifying $\frac{n^{[2k]}}{(2k)!}$ using Lemma (3.3), we get

$$\begin{aligned} n^{2m} &= \sum_{k=1}^{m} (2k)! \frac{n}{2k} \binom{n+k-1}{2k-1} T(2m, 2k) \\ &= \sum_{k=1}^{m} (2k-1)! n \binom{n+k-1}{2k-1} T(2m, 2k). \end{aligned}$$

By factoring out n, we obtain

$$n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k-1}{2k-1} T(2m, 2k).$$

By hockey stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ formula for sums of powers yields

$$\Sigma^1 n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k}{2k} T(2m, 2k).$$

The formula above matches the Proposition (5.1) precisely. Clearly, the formula for sums of odd powers can be generalized to multifold case, by applying hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ repeatedly

Proposition 5.2 (Knuth's odd powers sum multifold)**.** *For non-negative integers r, n, m*

$$\Sigma^{r+1} n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k+r}{2k+r} T(2m, 2k),$$

where $T(2m, 2k)$ are central factorial numbers of the second kind.

For example,

$$\Sigma^1 n^1 = \tbinom{n+1}{2} 1,$$

$$\Sigma^1 n^3 = \tbinom{n+2}{4} 6 + \tbinom{n+1}{2} 1,$$

$$\Sigma^1 n^5 = \tbinom{n+3}{6} 120 + \tbinom{n+2}{4} 30 + \tbinom{n+1}{2},$$

$$\Sigma^1 n^7 = \tbinom{n+4}{8} 5040 + \tbinom{n+3}{6} 1680 + \tbinom{n+2}{4} 126 + \tbinom{n+1}{2} 1.$$

While multifold sums of odd powers are,

$$\begin{aligned}
\Sigma^r n^1 &= \tbinom{n+0+r}{1+r}1, \\
\Sigma^r n^3 &= \tbinom{n+1+r}{3+r}6 + \tbinom{n+0+r}{1+r}1, \\
\Sigma^r n^5 &= \tbinom{n+2+r}{5+r}120 + \tbinom{n+1+r}{3+r}30 + \tbinom{n+0+r}{1+r}1, \\
\Sigma^r n^7 &= \tbinom{n+3+r}{7+r}5040 + \tbinom{n+3+r}{5+r}1680 + \tbinom{n+1+r}{3+r}126 + \tbinom{n+0+r}{1+r}1.
\end{aligned}$$

The coefficients $1, 6, 1, 120, 30, 1, \ldots$ is the sequence A303675 in the OEIS [16].

6. Generalized central factorial numbers of the second kind

Definition 6.1 (Generalized central factorial numbers of the second kind)**.** *For integers $0 \leq k \leq m$, and an arbitrary integer t, the generalized central factorial numbers of the second kind are defined by the relation*

$$n^m = \sum_{k=0}^{\infty} T_t(m,k) \cdot (n-t)^{[k]}.$$

It is also straightforward that

$$T_t(m,k) = \frac{\delta^k t^m}{k!},$$

by Newton's formula (3.2). For example,

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	0	1							
2	0	0	1						
3	0	$\frac{1}{4}$	0	1					
4	0	0	1	0	1				
5	0	$\frac{1}{16}$	0	$\frac{5}{2}$	0	1			
6	0	0	1	0	5	0	1		
7	0	$\frac{1}{64}$	0	$\frac{91}{16}$	0	$\frac{35}{4}$	0	1	
8	0	0	1	0	21	0	14	0	1

Table 1. Values of generalized central factorial numbers $T_0(n,k)$. See also the sequences A395862, and A370703 in the OEIS [16].

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	1	1							
2	1	2	1						
3	1	$\frac{13}{4}$	3	1					
4	1	5	7	4	1				
5	1	$\frac{121}{16}$	15	$\frac{25}{2}$	5	1			
6	1	$\frac{91}{8}$	31	35	20	6	1		
7	1	$\frac{1093}{64}$	63	$\frac{1491}{16}$	70	$\frac{119}{4}$	7	1	
8	1	$\frac{205}{8}$	127	$\frac{483}{2}$	231	126	42	8	1

Table 2. Values of generalized central factorial numbers $T_1(n,k)$. See also the sequences A395860, and A395861 in the OEIS [16].

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	2	1							
2	4	4	1						
3	8	$\frac{49}{4}$	6	1					
4	16	34	25	8	1				
5	32	$\frac{1441}{16}$	90	$\frac{85}{2}$	10	1			
6	64	$\frac{931}{4}$	301	190	65	12	1		
7	128	$\frac{37969}{64}$	966	$\frac{12411}{16}$	350	$\frac{371}{4}$	14	1	
8	256	$\frac{6001}{4}$	3025	3003	1701	588	126	16	1

Table 3. Values of generalized central factorial numbers $T_2(n, k)$. See also the sequences A394466, and A395314 in the OEIS [16].

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	3	1							
2	9	6	1						
3	27	$\frac{109}{4}$	9	1					
4	81	111	55	12	1				
5	243	$\frac{6841}{16}$	285	$\frac{185}{2}$	15	1			
6	729	$\frac{12753}{8}$	1351	585	140	18	1		
7	2187	$\frac{372709}{64}$	6069	$\frac{53011}{16}$	1050	$\frac{791}{4}$	21	1	
8	6561	$\frac{167943}{8}$	26335	$\frac{35049}{2}$	6951	1722	266	24	1

Table 4. Values of generalized central factorial numbers $T_3(n, k)$. See also the sequences A396559, and A395861 in the OEIS [16].

t	$T_t(2m, 2k)$	OEIS ID
0	$1, 0, 1, 0, 1, 1, 0, 1, 5, 1, 0, 1, 21, 14, 1, 0, 1, 85, 147, 30, 1, 0, 1, \cdots$	A269945
1	$1, 1, 1, 1, 7, 1, 1, 31, 20, 1, 1, 127, 231, 42, 1, 1, 511, 2290, 987, \cdots$	A394692
2	$1, 4, 1, 16, 25, 1, 64, 301, 65, 1, 256, 3025, 1701, 126, 1, 1024, \cdots$	A395456
3	$1, 9, 1, 81, 55, 1, 729, 1351, 140, 1, 6561, 26335, 6951, 266, 1, \cdots$	A395457

Table 5. Sequences of $T_t(2m, 2k)$.

Thus, we have the formula for sums of powers in terms of generalized central factorial numbers of the second kind

Proposition 6.2 (Generalized Central factorial sums)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^{m} \left[\binom{n - t + \frac{k}{2} + 1}{k+1} - \binom{1 - t + \frac{k}{2}}{k+1} + \binom{n - t + \frac{k}{2}}{k+1} - \binom{-t + \frac{k}{2}}{k+1} \right] k! T_t(m, k).$$

7. Double and triple sums of powers in terms of central differences

Proposition 7.1 (Double sums of powers decomposition)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\begin{aligned}\Sigma^2 n^m = \frac{1}{2} \sum_{k=0}^{m} \Bigg[&+ \binom{n - t + \frac{k}{2} + 2}{k+2} - \binom{2 - t + \frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{1 - t + \frac{k}{2}}{k+1} \Sigma^1 n^0 \\ &+ \binom{n - t + \frac{k}{2} + 1}{k+2} - \binom{1 - t + \frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{0 - t + \frac{k}{2}}{k+1} \Sigma^1 n^0 \Bigg] \delta^k t^m.\end{aligned}$$

For example,

Example 7.2 (Double sum of squares in central differences).

$$
\begin{aligned}
t=0: &\quad \Sigma^2 n^2 = \frac{t^2}{2}\,n(1+n) + \frac{\delta t^2}{2}\,\frac{n(2+3n+n^2)}{3} + \frac{\delta^2 t^2}{2}\,\frac{n(1+n)^2(2+n)}{12},\\
t=1: &\quad \Sigma^2 n^2 = \frac{t^2}{2}\,n(1+n) + \frac{\delta t^2}{2}\,\frac{n(-1+n^2)}{3} + \frac{\delta^2 t^2}{2}\,\frac{n^2(-1+n^2)}{12},\\
t=2: &\quad \Sigma^2 n^2 = \frac{t^2}{2}\,n(1+n) + \frac{\delta t^2}{2}\,\frac{n(-4-3n+n^2)}{3} + \frac{\delta^2 t^2}{2}\,\frac{n(10+5n-4n^2+n^3)}{12},\\
t=3: &\quad \Sigma^2 n^2 = \frac{t^2}{2}\,n(1+n) + \frac{\delta t^2}{2}\,\frac{n(-7-6n+n^2)}{3} + \frac{\delta^2 t^2}{2}\,\frac{n(32+23n-8n^2+n^3)}{12},\\
t=4: &\quad \Sigma^2 n^2 = \frac{t^2}{2}\,n(1+n) + \frac{\delta t^2}{2}\,\frac{n(-10-9n+n^2)}{3} + \frac{\delta^2 t^2}{2}\,\frac{n(66+53n-12n^2+n^3)}{12}.
\end{aligned}
$$

Proposition 7.3 (Triple sums of powers decomposition). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$
\begin{aligned}
\Sigma^3 n^m = \frac{1}{2}\sum_{k=0}^{m} \Bigg[&\\
&+ \binom{n-t+\frac{k}{2}+3}{k+3} - \binom{3-t+\frac{k}{2}}{k+3}\Sigma^0 n^0 - \binom{2-t+\frac{k}{2}}{k+2}\Sigma^1 n^0 - \binom{1-t+\frac{k}{2}}{k+1}\Sigma^2 n^0\\
&+ \binom{n-t+\frac{k}{2}+2}{k+3} - \binom{2-t+\frac{k}{2}}{k+3}\Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+2}\Sigma^1 n^0 - \binom{0-t+\frac{k}{2}}{k+1}\Sigma^2 n^0\Bigg]\delta^k t^m.
\end{aligned}
$$

For example,

Example 7.4 (Triple sum of squares in central differences).

$$
\begin{aligned}
t=0: &\quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(6+11n+6n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(18+45n+40n^2+15n^3+2n^4)}{120},\\
t=1: &\quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-2-n+2n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(-2-5n+5n^3+2n^4)}{120},\\
t=2: &\quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-10-13n-2n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(58+65n-5n^3+2n^4)}{120},\\
t=3: &\quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-18-25n-6n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(198+255n+40n^2-15n^3+2n^4)}{120},\\
t=4: &\quad \Sigma^3 n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-26-37n-10n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(418+565n+120n^2-25n^3+2n^4)}{120}.
\end{aligned}
$$

8. Multifold sums of powers in terms of central differences

Theorem 8.1 (Multifold sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\left\{\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s}\right]\Sigma^s n^0\right\}\delta^k t^m.$$

For example,

Example 8.2 (Examples of multifold sums)**.**

$$t=0: \quad \Sigma^r n^m = \tfrac{1}{2}\sum_{k=0}^{m}\left\{\tbinom{n+\frac{k}{2}+r}{k+r} + \tbinom{n+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\tbinom{r-s+\frac{k}{2}}{k+r-s} + \tbinom{r-1-s+\frac{k}{2}}{k+r-s}\right]\Sigma^s n^0\right\}\delta^k 0^m,$$

$$t=1: \quad \Sigma^r n^m = \tfrac{1}{2}\sum_{k=0}^{m}\left\{\tbinom{n-1+\frac{k}{2}+r}{k+r} + \tbinom{n-1+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\tbinom{r-s-1+\frac{k}{2}}{k+r-s} + \tbinom{r-2-s+\frac{k}{2}}{k+r-s}\right]\Sigma^s n^0\right\}\delta^k 1^m,$$

$$t=2: \quad \Sigma^r n^m = \tfrac{1}{2}\sum_{k=0}^{m}\left\{\tbinom{n-2+\frac{k}{2}+r}{k+r} + \tbinom{n-2+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\tbinom{r-s-2+\frac{k}{2}}{k+r-s} + \tbinom{r-3-s+\frac{k}{2}}{k+r-s}\right]\Sigma^s n^0\right\}\delta^k 2^m.$$

9. Multifold sums of powers in generalized central factorial numbers

Proposition 9.1 (Multifold central factorial sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\left\{\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s}\right]\Sigma^s n^0\right\}k!T_t(m,k).$$

For example,

Example 9.2 (Sums of squares in central factorial numbers)**.**

$$
\begin{aligned}
t=0: &\quad \Sigma^1 n^2 = n\cdot 0 + \tfrac{1}{2}n(1+n)\cdot 0 + \tfrac{1}{6}n(1+3n+2n^2)\cdot 1,\\
t=1: &\quad \Sigma^1 n^2 = n\cdot 1 + \tfrac{1}{2}n(n-1)\cdot 2 + \tfrac{1}{6}n(1-3n+2n^2)\cdot 1,\\
t=2: &\quad \Sigma^1 n^2 = n\cdot 4 + \tfrac{1}{2}n(n-3)\cdot 4 + \tfrac{1}{6}n(13-9n+2n^2)\cdot 1,\\
t=3: &\quad \Sigma^1 n^2 = n\cdot 9 + \tfrac{1}{2}n(n-5)\cdot 6 + \tfrac{1}{6}n(37-15n+2n^2)\cdot 1,\\
t=4: &\quad \Sigma^1 n^2 = n\cdot 16 + \tfrac{1}{2}n(n-7)\cdot 8 + \tfrac{1}{6}n(73-21n+2n^2)\cdot 1,\\
t=5: &\quad \Sigma^1 n^2 = n\cdot 25 + \tfrac{1}{2}n(n-9)\cdot 10 + \tfrac{1}{6}n(121-27n+2n^2)\cdot 1.
\end{aligned}
$$

From the example above, we may observe that

- For $t=0$ the coefficients $0,0,1$ correspond to the second row of the triangle (1).
- For $t=1$ the coefficients $1,2,1$ correspond to the second row of the triangle (2).
- For $t=2$ the coefficients $4,4,1$ correspond to the second row of the triangle (3).
- For $t=3$ the coefficients $9,6,1$ correspond to the second row of the triangle (4).

For example,

Example 9.3 (Sums of cubes in central factorial numbers)**.**

$$t = 0: \quad \Sigma^1 n^4 = n \cdot 0 + \tfrac{1}{2}n(1+n) \cdot 0 + \tfrac{1}{6}n(1+3n+2n^2) \cdot 1$$
$$+ \tfrac{1}{8}n(-1+n+4n^2+2n^3) \cdot 0 + \tfrac{1}{10}n(-2-5n+5n^3+2n^4) \cdot 1,$$
$$t = 1: \quad \Sigma^1 n^4 = n \cdot 1 + \tfrac{1}{2}n(n-1) \cdot 5 + \tfrac{1}{6}n(1-3n+2n^2) \cdot 7$$
$$+ \tfrac{1}{8}n(1+n-4n^2+2n^3) \cdot 4 + \tfrac{1}{10}n(-2+5n-5n^3+2n^4) \cdot 1,$$
$$t = 2: \quad \Sigma^1 n^4 = n \cdot 16 + \tfrac{1}{2}n(n-3) \cdot 34 + \tfrac{1}{6}n(13-9n+2n^2) \cdot 25$$
$$+ \tfrac{1}{8}n(-21+25n-12n^2+2n^3) \cdot 8$$
$$+ \tfrac{1}{10}n(18-45n+40n^2-15n^3+2n^4) \cdot 1,$$
$$t = 3: \quad \Sigma^1 n^4 = n \cdot 81 + \tfrac{1}{2}n(n-5) \cdot 111 + \tfrac{1}{6}n(37-15n+2n^2) \cdot 55$$
$$+ \tfrac{1}{8}n(-115+73n-20n^2+2n^3) \cdot 12$$
$$+ \tfrac{1}{10}n(298-275n+120n^2-25n^3+2n^4) \cdot 1.$$

From the example above, we may observe that

- For $t = 0$ the coefficients $0, 0, 1, 0, 1$ correspond to the fourth row of the triangle (1).
- For $t = 1$ the coefficients $1, 5, 7, 4, 1$ correspond to the fourth row of the triangle (2).
- For $t = 2$ the coefficients $16, 34, 25, 8, 1$ correspond to the fourth row of the triangle (3).
- For $t = 3$ the coefficients $81, 111, 55, 12, 1$ correspond to the fourth row of the triangle (4).

10. Multifold sums of powers in binomial form

Lemma 10.1 (Multifold sums of zero powers)**.** *For integers $r \geq 0$, and $n \geq 1$*

$$\Sigma^r n^0 = \binom{r+n-1}{r}.$$

Proof.

(1) Let $r = 0$, then we have $\Sigma^0 n^0 = 1$, by definition (2.1).

(2) Let $r=1$, then we have $\Sigma^1 n^0 = \sum_{k=1}^n 1 = \binom{n}{1}$.

(3) Let $r=2$, then we have $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \binom{n+1}{2}$.

(4) Let $r=3$, then we have $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$.

(5) Similarly, for arbitrary integer r, by hockey stick identity $\sum_{k=1}^n \binom{k}{r} = \binom{n+1}{r+1}$, yields $\Sigma^r n^0 = \sum_{k=1}^n \Sigma^{r-1} k^0 = \sum_{k=1}^n \binom{k+r-2}{r-1} = \binom{r+n-1}{r}$.

This completes the proof. □

Thus, by Lemma (10.1), and by Theorem (8.1) yields

Proposition 10.2 (Multifold binomial sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\left\{\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r}\right.$$
$$\left. - \sum_{s=0}^{r-1}\left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s}\right]\binom{s+n-1}{s}\right\}\delta^k t^m.$$

For example,

Example 10.3 (Examples of multifold binomial sums)**.**

$$t=0: \quad \Sigma^r n^m = \tfrac{1}{2}\sum_{k=0}^{m}\left\{\binom{n+\frac{k}{2}+r}{k+r} + \binom{n+\frac{k}{2}+r-1}{k+r}\right.$$
$$\left. - \sum_{s=0}^{r-1}\left[\binom{r-s+\frac{k}{2}}{k+r-s} + \binom{r-1-s+\frac{k}{2}}{k+r-s}\right]\binom{s+n-1}{s}\right\}\delta^k 0^m,$$

$$t=1: \quad \Sigma^r n^m = \tfrac{1}{2}\sum_{k=0}^{m}\left\{\binom{n-1+\frac{k}{2}+r}{k+r} + \binom{n-1+\frac{k}{2}+r-1}{k+r}\right.$$
$$\left. - \sum_{s=0}^{r-1}\left[\binom{r-s-1+\frac{k}{2}}{k+r-s} + \binom{r-2-s+\frac{k}{2}}{k+r-s}\right]\binom{s+n-1}{s}\right\}\delta^k 1^m,$$

$$t=2: \quad \Sigma^r n^m = \tfrac{1}{2}\sum_{k=0}^{m}\left\{\binom{n-2+\frac{k}{2}+r}{k+r} + \binom{n-2+\frac{k}{2}+r-1}{k+r}\right.$$
$$\left. - \sum_{s=0}^{r-1}\left[\binom{r-s-2+\frac{k}{2}}{k+r-s} + \binom{r-3-s+\frac{k}{2}}{k+r-s}\right]\binom{s+n-1}{s}\right\}\delta^k 2^m.$$

11. Framework for sums of powers

In this manuscript, we derive formulas for sums of powers by combining centered Newton's interpolation formula for powers (3.2) with hockey-stick identities (3.6), and (3.7). This approach can be generalized further. In particular, while using Newton's formulas

$$\begin{cases} f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^{[k]}}{k!}\delta^k f(a) = \sum_{k=0}^{\infty} \frac{x-a}{k}\binom{x-a+\frac{k}{2}-1}{k-1}\delta^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)_k}{k!}\Delta^k f(a) = \sum_{k=0}^{\infty} \binom{x-a}{k}\Delta^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^{(k)}}{k!}\nabla^k f(a) = \sum_{k=0}^{\infty} \binom{x-a+k-1}{k}\nabla^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!}\frac{\mathrm{d}^k f(a)}{\mathrm{d}x^k}. \end{cases} \tag{1}$$

one may replace linear difference operators δ, Δ, ∇ with their non-linear dynamic analogs, allowing more generic approach for sums of powers. It is indeed so remarkable that Taylor's series $f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!}\frac{\mathrm{d}^k f(a)}{\mathrm{d}x^k}$ is a continuous analogue of Newton's formula! Back to the topic, such non-linear difference operators arise naturally in the theory of dynamic equations on time scales, developed by Bohner and Peterson in [17]. Thus, the discrete change $x+h$ of a function $f(x+h)$ is replaced by function $g(h)$ which implies that dynamic difference operator is

$$D_g f = \frac{f(g(x)) - f(x)}{g(x) - x}. \tag{2}$$

A well-known example is Jackson's q-difference [18]

$$D_q f(x) = \frac{f(qx) - f(x)}{qx - x},$$

where $g(x) = qx$. In particular, for Newton's formula in linear differences δ, Δ, ∇, their respective falling, rising, and central factorials are related to binomial coefficients via

$$\begin{cases} \frac{(x)_n}{n!} &= \frac{1}{n!}x(x-1)(x-2)\cdots(x-n+1) = \binom{x}{n}, \\ \frac{x^{(n)}}{n!} &= \frac{1}{n!}x(x+1)(x+2)\cdots(x+n-1) = \binom{x+n-1}{n}, \\ \frac{x^{[n]}}{n!} &= \frac{1}{n!}\left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = \frac{x}{n}\binom{x+\frac{n}{2}-1}{n-1}. \end{cases} \tag{3}$$

Motivated by this viewpoint, we can see that dynamic version of Newton's formula is

$$f(x) = \sum_{k=0}^{\infty} \frac{(x \mid \alpha, \beta)_k}{k!} \cdot D_g f(a), \tag{4}$$

where a is an arbitrary integer, $D_g f(a)$ is dynamic difference operator, and $(x \mid \alpha, \beta)_n$ is generalized factorial which uses the combination of Nörlund's notation [19], and definition given by Steffensen [9]

Definition 11.1 (Generalized factorial)**.** *For non-negative integers x, n, α, β*

$$(x \mid \alpha, \beta)_n = x(x - n\alpha - \beta)(x - n\alpha - 2\beta) \cdots (x - n\alpha - n - I\beta).$$

For instance,

$$\begin{cases} (x)_n & = (x \mid \alpha, \beta)_n, \quad \alpha = 0, \quad \beta = 1, \quad \text{(forward Newton's formula)} \\ x^{(n)} & = (x \mid \alpha, \beta)_n, \quad \alpha = 0, \quad \beta = -1, \quad \text{(backward Newton's formula)} \\ x^{[n]} & = (x \mid \alpha, \beta)_n, \quad \alpha = \frac{1}{2}, \quad \beta = 1, \quad \text{(central Newton's formula)} \\ x^n & = (x \mid \alpha, \beta)_n, \quad \alpha = 0, \quad \beta = 0, \quad \text{(Taylor's formula)} \end{cases}$$

More precisely, the algorithm for closed formulas for sums of powers is

Algorithm 11.2 (Algorithm for closed formulas)**.**

(1) Define the function $f(x) = x^m$.

(2) Pick generalized difference operator $D_g f(x)$, see (2).

(3) Derive generalized Newton's formula (4) in terms of $D_g f(x)$, for example (1).

(4) Find an identity that connects respective factorials $(x \mid \alpha, \beta)_n$ with binomial coefficients of the form $\binom{x}{k}$, so that upper index is expression in x, for example (3).

(5) Apply hockey stick identities (3.6), and (3.7) to derive closed form for sums of powers $\Sigma^1 n^m$.

(6) Apply identities (3.6), and (3.7) r-times to reach r-fold closed form for sums of powers $\Sigma^r n^m$.

The algorithm above has already been utilized in practice. For instance, the work [20] provides remarkable formulas for sums of powers, based on forward difference Newton's formula,

Proposition 11.3. *For integers $r \geq 0$, and $n \geq 0$, and $m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r n^m = \sum_{j=0}^{m} \left[\binom{n-t+r}{j+r} + \sum_{s=1}^{r} (-1)^{j+s-1} \binom{j+t-1}{j+s} \binom{r-s+n-1}{r-s} \right] \Delta^j t^m.$$

Also, in terms of Eulerian numbers $\left\langle {n \atop k} \right\rangle$

Proposition 11.4. *For integers $r \geq 0$, and $n \geq 0$, and $m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r n^m = \sum_{j=0}^{m} \sum_{k=0}^{m} \left[\binom{n-t+r}{j+r} - \sum_{s=1}^{r} \binom{s-t}{j+s} \binom{r-s+n-1}{r-s} \right] \binom{t+k}{m-j} \left\langle {m \atop k} \right\rangle.$$

The work entitled *Sums of powers of integers and generalized Stirling numbers of the second kind* [21] discusses closed formulas for sums of powers in terms of generalized Stirling numbers of the second kind, which originates from forward Newton's formula for powers, around arbitrary integer point t. It is quite remarkable that both [21], and [20] reached the same result independently. In [21] we get

Proposition 11.5 (Cereceda (2022))**.**

$$\Sigma^1 n^k = \sum_{j=0}^{k} j! \left[\binom{n+1-r}{j+1} + (-1)^j \binom{r+j-1}{j+1} \right] \left\{ {k \atop j} \right\}_r, \tag{5}$$

$$\Sigma^1 n^k = \sum_{j=0}^{k} j! \left[\binom{n+1-r}{j+1} + \binom{r+1}{j+1} \right] \left\{ {k \atop j} \right\}_{-r}. \tag{6}$$

Formula (5) is a special case of proposition [20, Prop. (1.9)]

Proposition 11.6. *For integers $n \geq 0$, and $m \geq 0$, and an arbitrary integer t*

$$\Sigma^1 n^m = \sum_{j=0}^{m} \Delta^j t^m \left[\binom{n-t+1}{j+1} + (-1)^j \binom{j+t-1}{j+1} \right].$$

in terms of generalized Stirling numbers $\left\{ {k \atop j} \right\}_r$. This implies that finite differences of powers can be expressed in terms of generalized Stirling numbers of the second kind, $\Delta^j t^m = j!\left\{ {m \atop j} \right\}_t$. Formula (6) is a special case of proposition [20, Prop. (1.8)]

Proposition 11.7. *For integers $n \ge 0$, and $m \ge 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \sum_{j=0}^{m} \Delta^j t^m \left[\binom{n-t+1}{j+1} - \binom{1-t}{j+1} \right].$$

with setting $t = r$. Moving to backward Newton's formula, the work [22] utilizes algorithm (11.2) in terms of backward differences, providing another remarkable formula

Proposition 11.8. *For non-negative integers r, n, m and an arbitrary integer t,*

$$\Sigma^{r+1} n^m = \sum_{j=0}^{m} \left[\binom{j-t+n+r}{j+r+1} + \sum_{s=0}^{r} (-1)^{j+s} \binom{t}{j+s+1} \binom{r-s+n-1}{r-s} \right] \nabla^j t^m.$$

Finally, the work [23] explores Newton's formula in central differences, allowing it to be applied with algorithm (11.2). Essentially, it achieves similar formulas for sums of powers as current paper, but using different binomial identities to reach the closed form. For instance,

Proposition 11.9. *For integers $n \ge 0$, $m \ge 0$, and an arbitrary integer t*

$$\Sigma^r n^m = \binom{r+n-1}{r} t^m + \sum_{k=1}^{m} \frac{\delta^k t^m}{2} \Bigg\{ \left[\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right] - \sum_{s=0}^{r-1} \left[\binom{-t+\frac{k}{2}+r-s}{k+r-s} + \binom{-t+\frac{k}{2}+r-s-1}{k+r-s} \right] \binom{s+n-1}{s} \Bigg\}.$$

12. Conclusions

This manuscript reviews historical development of the sums of powers problem, briefly tracing its evolution from Archimedes (287-212 BC) to Donald Knuth's work *Johann Faulhaber and sums of powers (1993)*, highlighting the transition from purely arithmetic methods to modern enumerative and algebraic-combinatorial approaches. The main results establish an algorithm (11.2) for deriving closed formulas for multifold sums of powers of integers by combining variations of Newton's interpolation formula with hockey-stick family identities for binomial coefficients. Using the algorithm, some remarkable formulas were obtained, closed formula for sums of powers (4.1), formula for $r-$fold sums of powers (8.1), formula for $r-$fold sums of powers in pure binomial view (10.2). In addition, we have shown that famous Knuth's formula for sums of odd powers (5.1) originates from Newton's interpolation formula (3.2), thus generalizing it for $r-$fold case (5.2).

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Mathematica programs

Use this `GitHub` repository to validate the results using Mathematica programs.

Mathematica Function	**Validates / Prints**
`MultifoldSumOfPowersRecurrence[r, n, m]`	Computes $\Sigma^r n^m$, (2.1)
`CentralFactorial[n, k]`	Computes $n^{[k]}$, (2.3)
`ValidateCentralFactorialsBinomialDecomposition.txt`	Validates Lem. (3.4)
`ValidateNewtonsFormulaForPowers.txt`	Validates Lem. (3.5)
`ValidateCenteredHockeyStickIdentity.txt`	Validates Lem. (3.7)
`ValidateOrdinarySumsOfPowersDecomposition.txt`	Validates Prop. (4.1)
`ValidateMultifoldSumsKnuthFormula.txt`	Validates Prop. (5.2)
`ValidateDoubleSumsOfPowersDecomposition.txt`	Validates Prop. (7.1)
`ValidateTripleSumsOfPowersDecomposition.txt`	Validates Prop. (7.3)
`ValidateMultifoldSumsOfPowersTheorem.txt`	Validates Thm. (8.1)
`ValidateMultifoldSumsOfZeroPowersLemma.txt`	Validates Lem. (10.1)
`ValidateMultifoldBinomialSumsOfPowersProposition.txt`	Validates Prop. (10.2)

Metadata

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